Bt603 current final paper fall 2020

BT603 Current paper dated 11-09-2020

1. **Constitutive enzyme**an enzyme that is synthesized continually regardless of growth conditions.
2. **Phenyl acetic acid is still the most widely used precursor in penicillin production.**
3. **Many types of antifoam are of low solubility so they need carriers.**
4. **Independent variable (nutrient, antifoam, pH and temperature.**
5. **When more than five independent variables are to be investigated, the Plackett-Burman design may be used to find the most important variables in a system, which are then optimized in further studies (Plackett and Burman, 1946).**
6. **The solubility of oxygen in pure water is 8 mg/L at 4oC.**
7. **During lyophilization, a suspension of microbes is quickly frozen at temperatures ranging from -54°C to -72°C,**
8. **Pilot- and industrial-scale fermentations are normally carried out in stirred, aerated vessels(fermenters).**
9. **Volumetric mass-transfer coefficient. The units of *KLa*, are reciprocal time (h-1).**
10. **Auxotrophy is the inability of an organism to synthesize a particular organic compound required for its growth.**
11. **The most common cause of foaming is due to proteins.**
12. **yeasts such as *Hansenula polymorpha* and *Pichiapastoris.***
13. Fermentation can beclassified into differenttypes on the7 basis.
14. **There are five major groups of commerciallyimportant fermentations.**
15. **1900-1940** Batch + ContinuousSystem.
16. **Detailed investigation is needed to establish the mostsuitable medium.**
17. **Selective media are designed to suppress the growthof unwanted bacteria and encourage the growth ofthe desired microbes.**
18. ***Barley malt*Reducing sugars3-4 moles.**
19. **Stoichiometric equations describing oxygen demand in afermentation C=(A/Y)—B.**
20. **Azotobacter sp at 30C is 0.018.**
21. **Alkalophiles are obtained from the high salinity soils.**
22. **Metabolites are of three types.**
23. SSF is also called as **Koji Fermentation.**
24. Antifoams **Should have no effect on oxygen transfer.**
25. Crude media is made up of unrefined agricultural products e.g. containing barley.
26. **Parton and Willis (1990) advocated the use of the carbon dioxide production rate (CPR) as atransfer criterion which requires analysis of the fermenter effluent air.**
27. **(NCTC) abbreviation of? National Collection of Type cultures.**
28. **Production of 100 grams of biomass from hydrocarbon requires approximately three times theamount of oxygen to produce the same amount of biomass from carbohydrate.**
29. **Antifoams are surfaceactive agents, reducingthe surface tension inthe foams anddestabilizing protein films.**
30. **Cellular yield coefficient of n-alkanes 1.03**

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1. What is the role of inhibitor in fermentation medium? 2 marks
2. What is random mutant genesis? 2 marks
3. Phonotypic characteristics of microorganisms? 3 marks
4. Name the three ways of introducing the air in fermenter? 3 marks
5. What is Replica plating? 3 marks
6. Write a note on molasses? 5
7. Explain biological and physical problems due to the formation of
8. What is Replica Plating Method? (5)
9. Technique of microbial culture storage/preservation explains it? 10 marks

